WEEK 5 REPORT

Topic: DAX

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Internship Domain: Data Analytics / Data Science
Tool Used: Power BI - DAX Query editor
Focus: Understanding and implementing DAX functions

Week 6 — DAX Queries (Power BI)

1. What is DAX?

DAX (Data Analysis Expressions) is a formula language used in Power BI, Excel Power Pivot, and SQL Server Analysis Services. It helps create custom calculations such as measures, calculated columns, and calculated tables.

2. Why is DAX Used in Power BI?

- Advanced calculations beyond simple aggregation
- Dynamic measures based on filters and slicers
- Business logic like growth %, KPIs, ratios
- Time intelligence (YoY, MoM, YTD)
- Better analytics and reporting

3. Advantages of Using DAX

- Powerful and flexible
- Context-aware calculations
- Efficient with large datasets
- Reusable measures
- Interactive dashboards

4. Terminologies in DAX

Measures

Measures are dynamic calculations used in visuals.
Example: Total Sales = SUM(Sales[Amount])

Calculated Columns

Calculated columns are computed row by row and stored in the model.
Example: Profit = Sales[Revenue] - Sales[Cost]

Sigma (Aggregation Functions)

Aggregation functions summarize data such as SUM(), AVERAGE(), COUNT(), MIN(), MAX().

5. Implicit vs Explicit Measures

Implicit Measures

Automatically created when dragging numeric fields into visuals.

Explicit Measures

Manually created using DAX formulas and reusable.
Example: Total Sales = SUM(Sales[Amount])

6. Conditions in DAX

Conditions apply logic using IF or SWITCH.
Example: Status = IF(Sales[Amount] > 1000, "High", "Low")

7. Context in DAX

Row Context: Calculation per row.
Filter Context: Changes based on filters and slicers.